

Restriction Enzyme Digestion

Overview

Linearize purified plasmids with restriction enzymes to verify their sizes.

A. Materials/Components

- a. Target DNA (plasmid of interest)
- b. Restriction Enzyme
- c. 10X CutSmart Buffer
- d. Sterile ddH₂O

B. Procedure

1. Set up following reaction on ice

$$\text{Volume of DNA } (\mu\text{l}) = \frac{\text{desired DNA mass (ng)}}{\text{DNA concentration (ng}/\mu\text{l})}$$

*desired DNA mass = 500ng

2. Calculate the reaction formula.

Reagent	Final con. or amount	1 Reaction
10X CutSmart buffer	1X	2 μl
Restriction enzyme	20 units/ μl	0.5 μl
DNA Template	500 ng	=500 (ng) / Plasmid conc.
ddH ₂ O	Top up to 20 μl	=20 - (sum of everything else)

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